Mathematics

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1 Linear Algebra

Linear algebra is central to almost all areas of mathematics. A good grasp on the subject is required for a solid foundation on higher level mathematics. References from Garcia and Horn (2017) and Axler (2024).

1.1 Vector Spaces

Definition 1.1.1. *Vector Space*
Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then $\mathcal{V}$ is a vector space over a field $\mathbb{F}$ if the following are satisfied:

i. Identities: $\exists\ \mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$, and $\exists\ \mathbf{1} \in \mathcal{V}$ such that $\mathbf{1u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$
ii. Commutativity: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}\ \forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
iii. Associativity: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ and $(ab)\mathbf{u} = a(b\mathbf{u})\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$
iv. Distributive: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ and $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
v. Additive Inverse: $\forall\ \mathbf{u} \in \mathcal{V}\ \exists$ unique vector $-\mathbf{u} \in \mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
vi. Multiplicative Inverse: $\forall\ \mathbf{u} \in \mathcal{V}$ with $\mathbf{u} \neq \mathbf{0}\ \exists$ unique $\mathbf{w} \in \mathcal{V}$ such that $\mathbf{uw} = 1$

Remark 1.1.2. *Properties of Vector Spaces*
Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$.

i. A vector space has a unique additive identity
ii. Every element in a vector space has a unique additive inverse
iii. $\mathbf{0v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$
iv. $a\mathbf{0} = \mathbf{0}$ for every $a \in \mathbb{F}$
v. $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$

Definition 1.1.3. *Subspace*
A subspace of an $\mathbb{F}$-vector space $\mathcal{V}$ is a subset $\mathcal{U}$ of $\mathcal{V}$ that is an $\mathbb{F}$-vector space with the same vector addition and scalar multiplication operations as in $\mathcal{V}$. To check $\mathcal{U}$ satisfies the following:

i. Additive Identity: $\mathbf{0} \in \mathcal{U}$
ii. Closure Under Addition: $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ implies $\mathbf{u} + \mathbf{v} \in \mathcal{U}$
iii. Closed Under Scalar Multiplication: $a \in \mathbb{F}$ and $\mathbf{u} \in \mathcal{U}$ implies $a\mathbf{u} \in \mathcal{U}$.

References

Axler, S. (2024). *Linear Algebra Done Right* (4 ed.). Undergraduate Texts in Mathematics. Cham: Springer.

Garcia, S. R. and R. A. Horn (2017). *A Second Course in Linear Algebra*. Cambridge: Cambridge University Press.